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Department of Computer Science and Engineering

Unit – 4 Network Layer and Internet Protocol

4.1 Overview of Network Layer

4.2 What's Inside a Router?

4.3 Switching

4.4 The Internet Protocol (IP)

- Datagram format
- Fragmentation
- IPv4 addressing
- NAT

4.5 Introduction to Network Layer Protocols

- DHCP
- ICMP

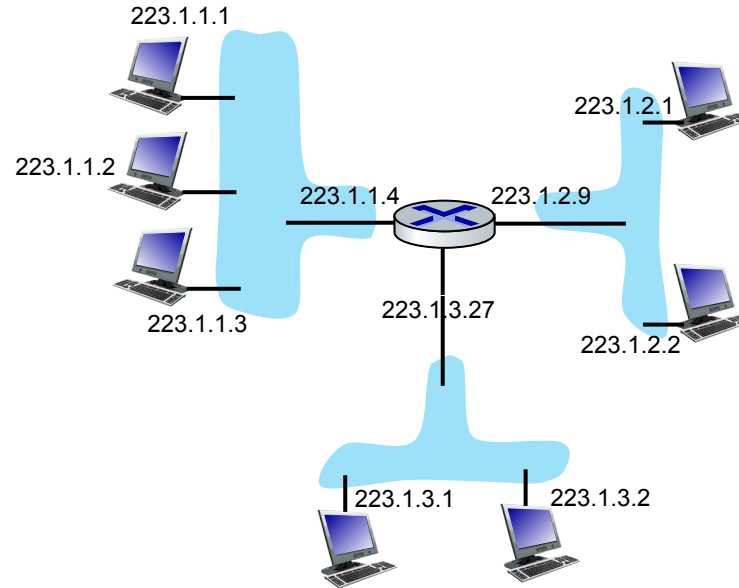
4.6 IPv6 Addressing

4.7 Introduction to Routing Algorithms

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IP addressing: Introduction

- **IP address:** 32-bit unique ID associated with each host or router *interface*
- **interface:** connection between host/router and physical link
 - router's typically have multiple interfaces
 - host typically has one or two interfaces (e.g., wired Ethernet, wireless 802.11)



dotted-decimal IP address notation:

223.1.1.1 = 11011111 00000001 00000001 00000001
223 1 1 1

Change the following IPv4 addresses from binary notation to dotted-decimal notation.

a. 10000001 00001011 00001011 11101111

b. 11000001 10000011 00011011 11111111

Solution:

We replace each group of 8 bits with its equivalent decimal number and add dots for separation.

a. 129.11.11.239

b. 193.131.27.255



Change the following IPv4 addresses from dotted-decimal notation to binary notation.

a. 111.56.45.78

b. 221.34.7.82

Solution:

We replace each decimal number with its binary equivalent.

a. 01101111 00111000 00101101 01001110

b. 11011101 00100010 00000111 01010010



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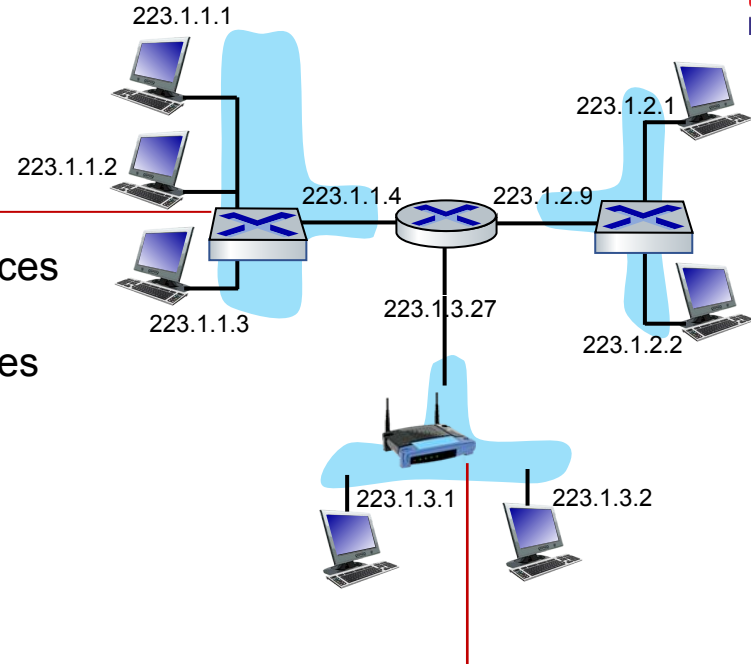
IP addressing: Introduction

Q: how are interfaces
actually connected?

A: we'll learn about that in
chapters 6, 7

A: wired
Ethernet interfaces
connected by
Ethernet switches

For now: don't need to worry
about how one interface is
connected to another (with no
intervening router)



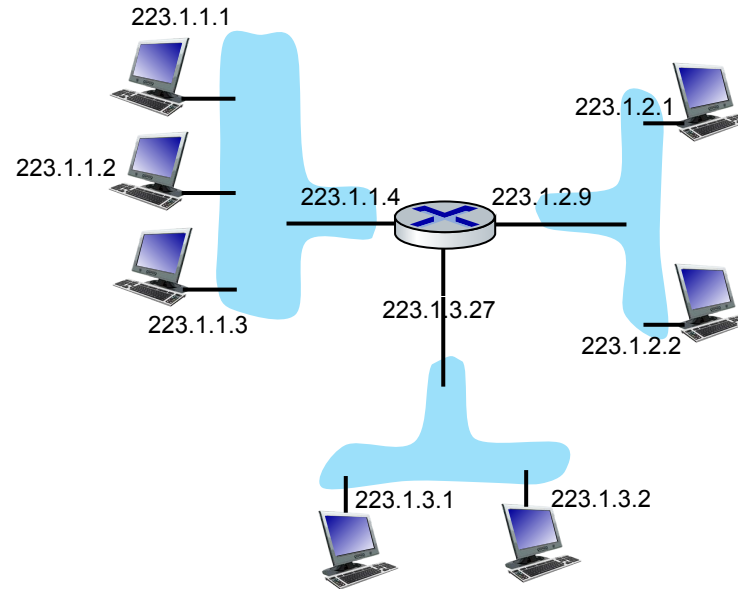
A: wireless WiFi interfaces
connected by WiFi base station

▪ *What's a subnet ?*

- device interfaces that can physically reach each other **without passing through an intervening router**

▪ IP addresses have structure:

- **subnet part**: devices in same subnet have common high order bits
- **host part**: **remaining** low order bits

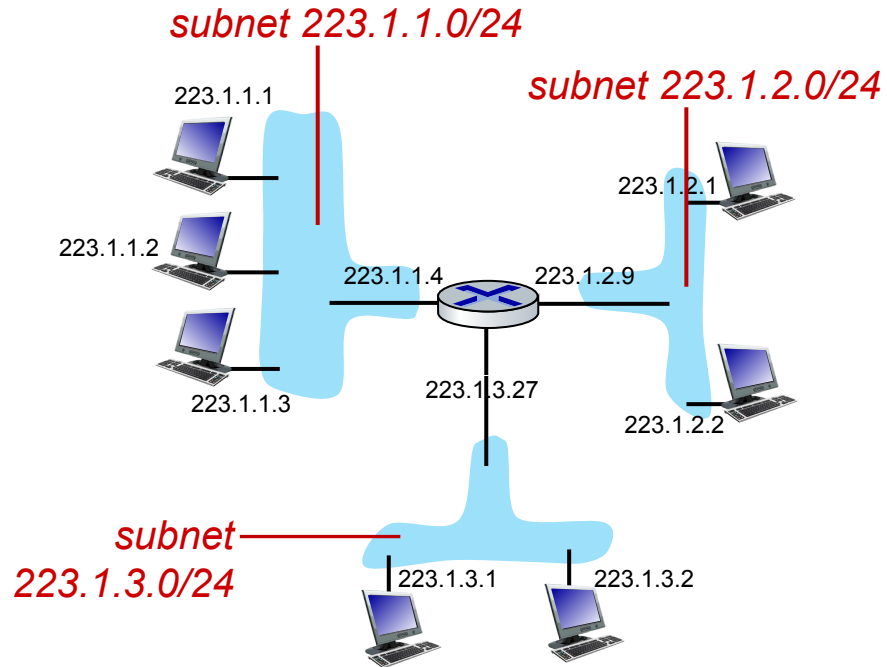


network consisting of 3 subnets

A portion of a network that shares a particular subnet address.

Recipe for defining subnets:

- detach each interface from its host or router, creating “islands” of isolated networks
- each isolated network is called a *subnet*



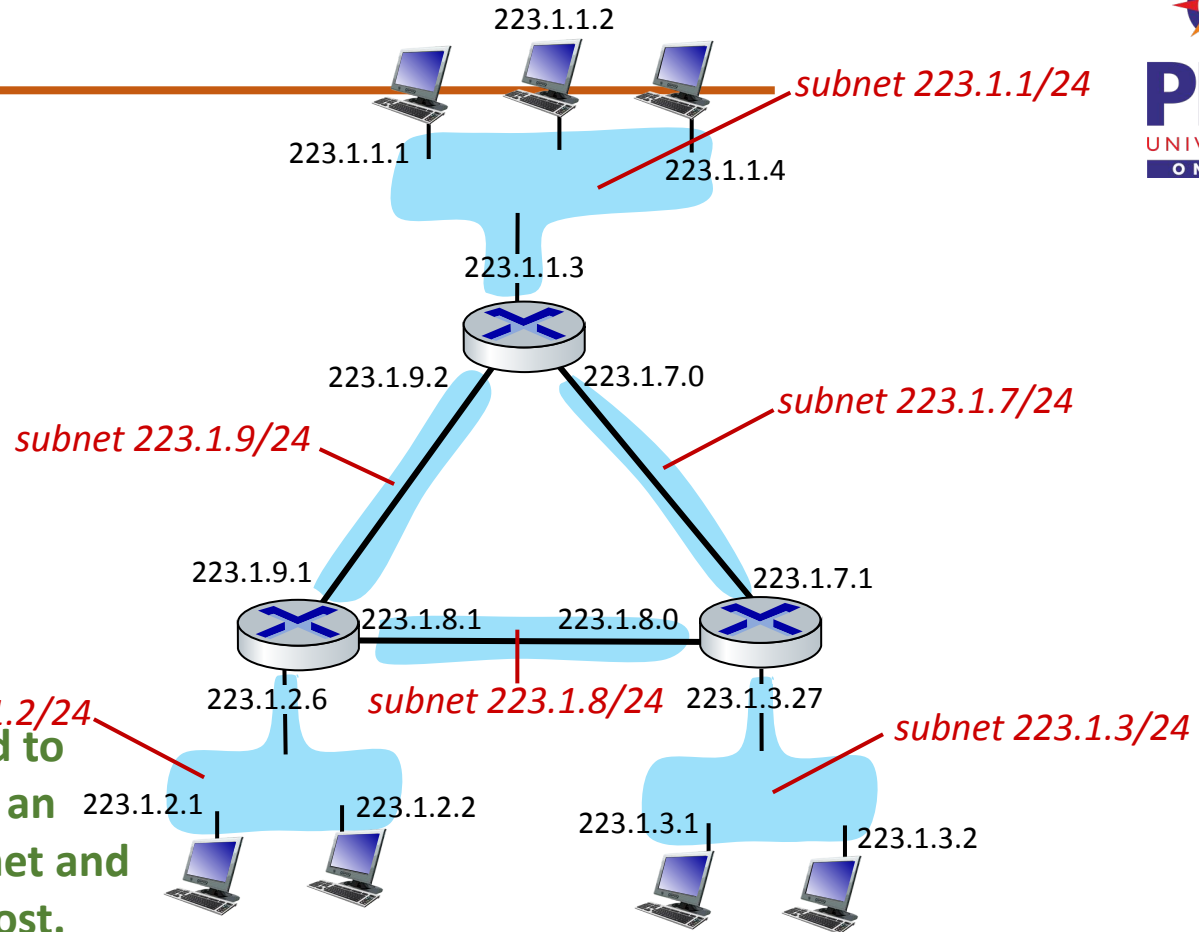
subnet mask: /24

(high-order 24 bits: subnet part of IP address)

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Subnets

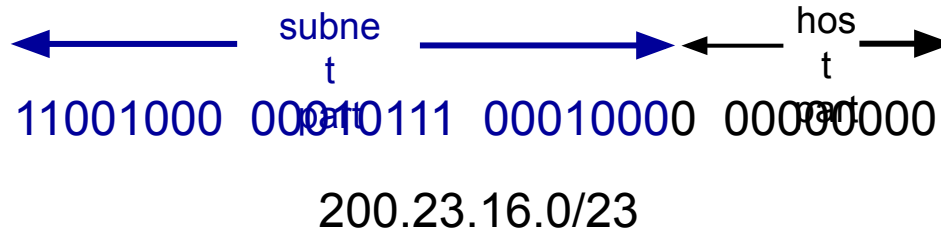
- where are the subnets?
- what are the /24 subnet addresses?



A 32-bit combination used to describe which portion of an address refers to the subnet and which part refers to the host.

CIDR: Classless InterDomain Routing (pronounced "cider")

- subnet portion of address of arbitrary length
- address format: **a.b.c.d/x**, where x is # bits in subnet portion of address



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IP addressing: CIDR



CIDR Block Prefix	# of Host Addresses (inclusive of network and broadcast-id)
-------------------	---

/30	4 hosts (valid host would be 2)
/29	8 hosts (valid host would be 6)
/28	16 hosts (valid host would be 14)
/27	32 hosts
/26	64 hosts
/25	128 hosts
/24	256 hosts
/23	512 hosts
/22	1,024 hosts
/21	2,048 hosts
/20	4,096 hosts
/19	8,192 hosts
/18	16,384 hosts
/17	32,768 hosts
/16	65,536 hosts
/15	131,072 hosts
/14	262,144 hosts
/13	524,288 hosts
/12	1,048,576 hosts and so on.....

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IP addressing: Why not Classful Addressing?



- Network portion - 8, 16, or 24 bits in length – known as **Class A, B and C** networks respectively
- A class **C (/24)** - accommodate only up to $2^8 - 2 = 254$ **hosts** (two of the $2^8 = 256$ addresses are reserved for special use)
- A class **B (/16)** subnet, supports up to **65,534 hosts**
- A class **A (/8)** subnet, up to **16,777,214 hosts**

too small for many organizations

too large, poor utilization

1

$2^n = \text{Number of Network}$

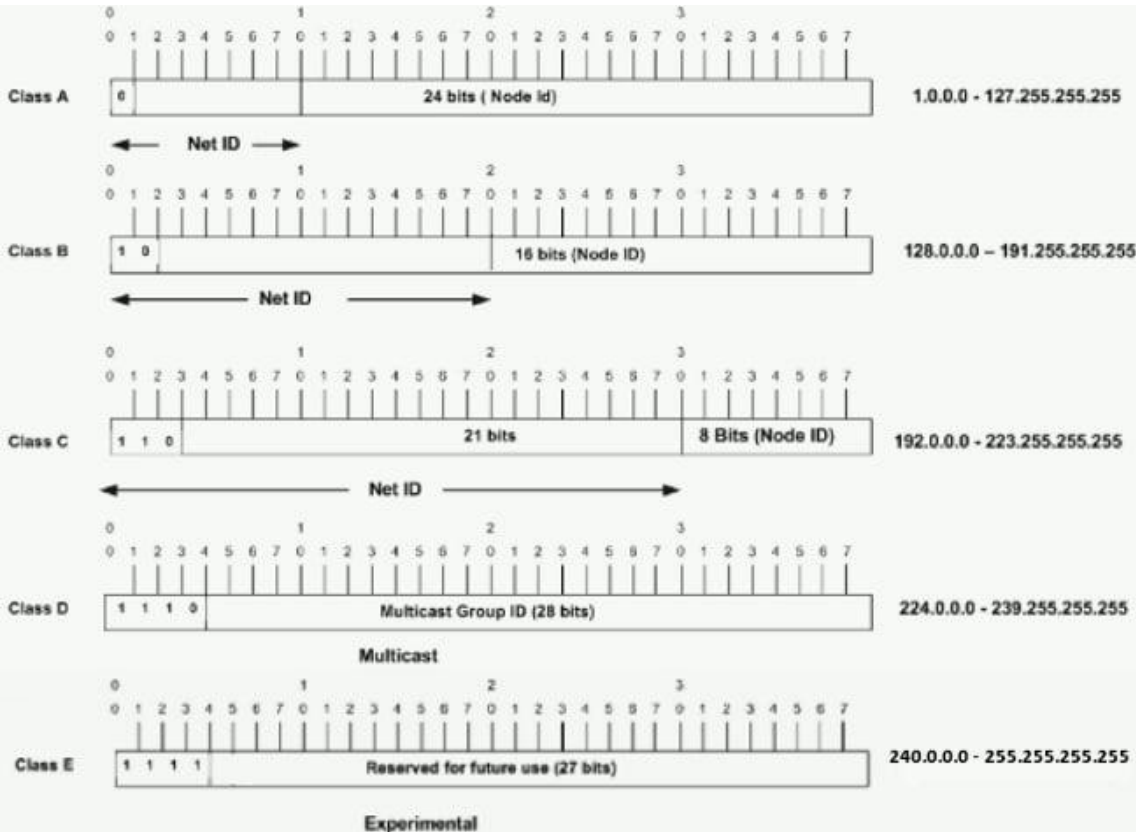
2

$2^h - 2 = \text{Number of Host}$

Classful addressing, which is almost obsolete, is replaced with classless addressing.

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IP addressing: Classful Addressing



Class A



Class B



Class C



Class D



Class E

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IP addressing: Classful Addressing



Class	HOB	NET ID Bits	Host ID Bits	No of Networks	Host Per Network	Start Address	End Address
Class A	0	8	24	$2^7=128$	$2^{24}=16,777,216$	0.0.0.0	127.255.255.255
Class B	10	16	16	$2^{14}=16,384$	$2^{16}=65,536$	128.0.0.0	191.255.255.255
Class C	110	24	8	$2^{21}=2,097,152$	$2^8=256$	192.0.0.0	223.255.255.255
Class D	1110	-	-	-	-	224.0.0.0	239.255.255.255
Class E	1111	-	-	-	-	240.0.0.0	255.255.255.255

Class D: multicast, Class E: reserved (experimental)

Broadcast Address: 255.255.255.255

Network Masks

Class A: 255.0.0.0

Class B: 255.255.0.0

Class C: 255.255.255.0

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IP addressing: Classful Subnet Masks

Class	Binary	Dotted-Decimal	CIDR Notation
A	11111111 00000000 00000000 00000000	255.0.0.0	/8
B	11111111 11111111 00000000 00000000	255.255.0.0	/16
C	11111111 11111111 11111111 00000000	255.255.255.0	/24

Q: how does *network* get subnet part of IP address?

A: gets allocated portion of its provider ISP's address space

ISP's block 11001000 00010111 00010000 00000000 200.23.16.0/20

ISP can then allocate out its address space in 8 blocks:

Organization 0	<u>11001000 00010111 00010000</u>	00000000	200.23.16.0/23
Organization 1	<u>11001000 00010111 00010010</u>	00000000	200.23.18.0/23
Organization 2	<u>11001000 00010111 00010100</u>	00000000	200.23.20.0/23
...			
Organization 7	<u>11001000 00010111 00011110</u>	00000000	200.23.30.0/23

Find the class of each address.

- a. 00000001 00001011 00001011 11101111
- b. 11000001 10000011 00011011 11111111
- c. 14.23.120.8
- d. 252.5.15.111

Solution

- a. The first bit is 0. This is a class A address.
- b. The first 2 bits are 1; the third bit is 0. This is a class C address.
- c. The first byte is 14; the class is A.
- d. The first byte is 252; the class is E.



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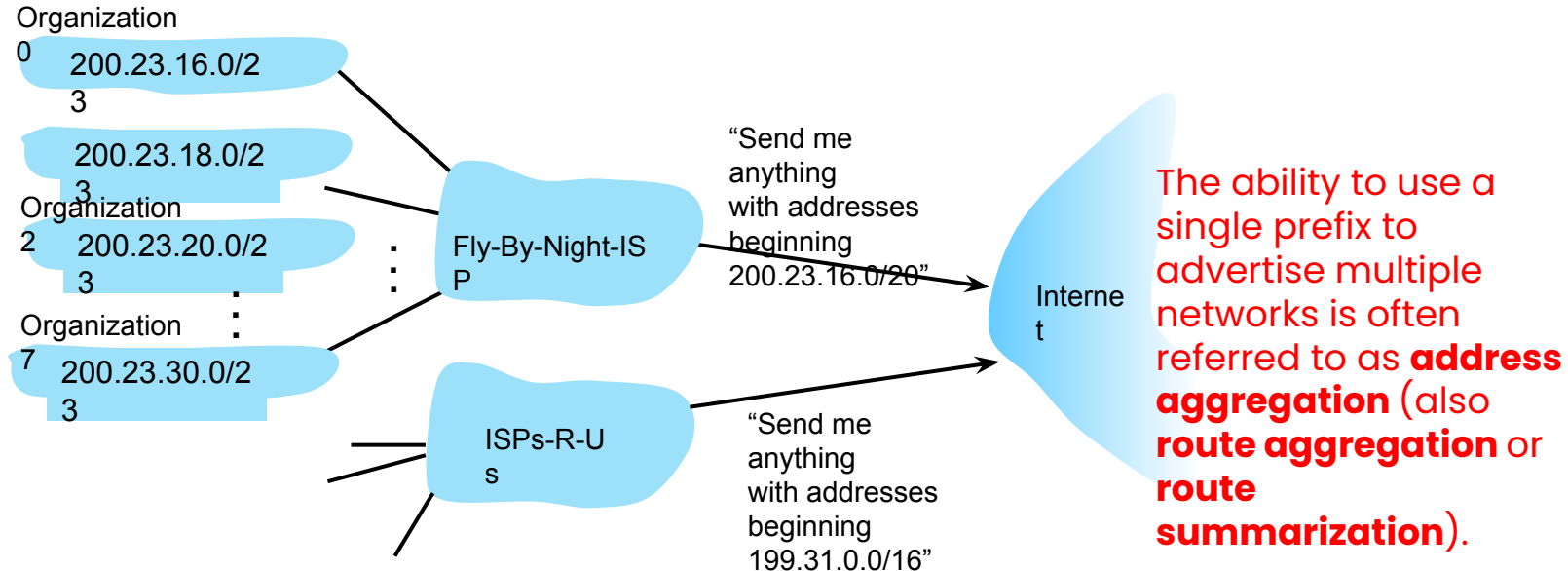
IP addressing: CIDR – Prefix Lengths



<i>/n</i>	<i>Mask</i>	<i>/n</i>	<i>Mask</i>	<i>/n</i>	<i>Mask</i>	<i>/n</i>	<i>Mask</i>
/1	128.0.0.0	/9	255.128.0.0	/17	255.255.128.0	/25	255.255.255.128
/2	192.0.0.0	/10	255.192.0.0	/18	255.255.192.0	/26	255.255.255.192
/3	224.0.0.0	/11	255.224.0.0	/19	255.255.224.0	/27	255.255.255.224
/4	240.0.0.0	/12	255.240.0.0	/20	255.255.240.0	/28	255.255.255.240
/5	248.0.0.0	/13	255.248.0.0	/21	255.255.248.0	/29	255.255.255.248
/6	252.0.0.0	/14	255.252.0.0	/22	255.255.252.0	/30	255.255.255.252
/7	254.0.0.0	/15	255.254.0.0	/23	255.255.254.0	/31	255.255.255.254
/8	255.0.0.0	/16	255.255.0.0	/24	255.255.255.0	/32	255.255.255.255

The addresses in color are the default masks for classes A, B, and C.
Thus, classful addressing is a special case of classless addressing.

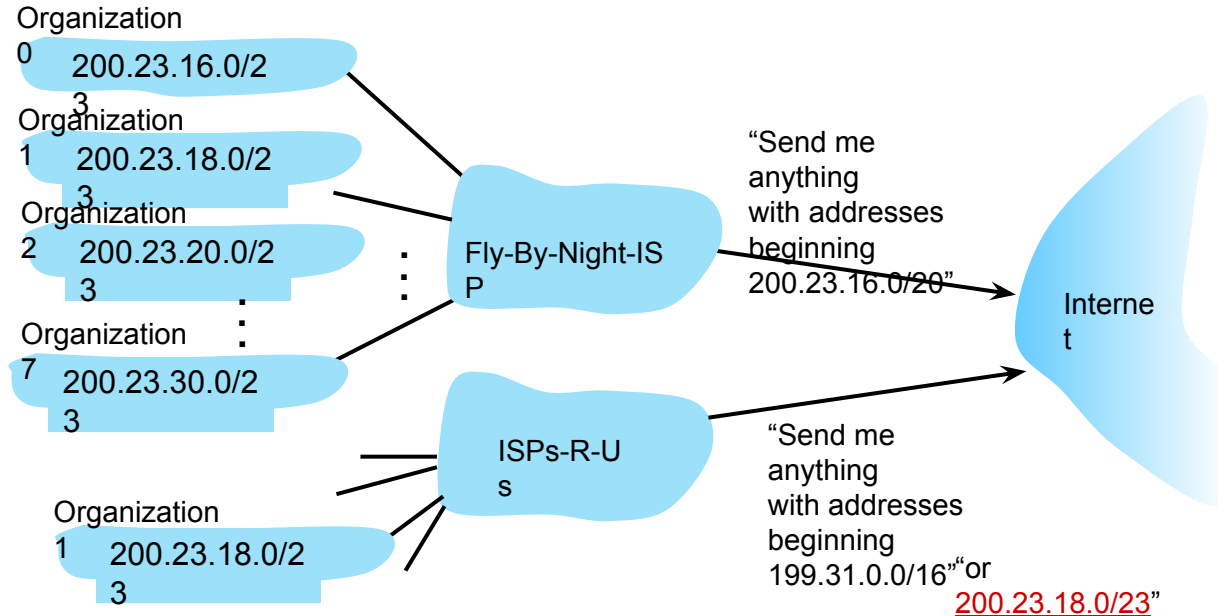
hierarchical addressing allows efficient advertisement of routing information:



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Hierarchical addressing: more specific routes

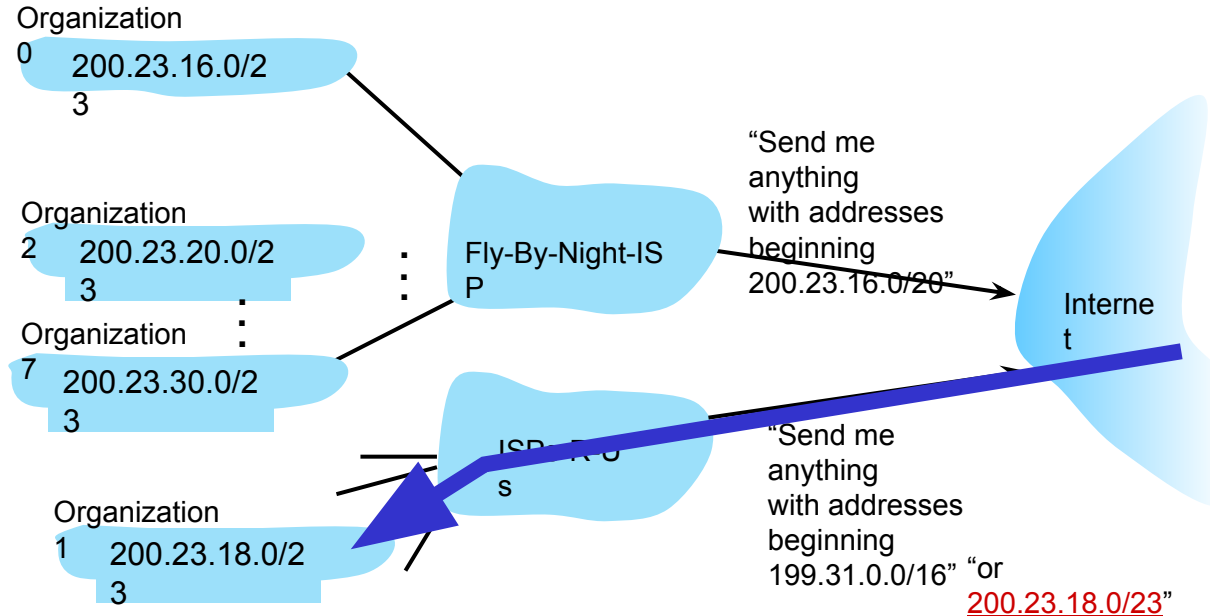
- Organization 1 moves from Fly-By-Night-ISP to ISPs-R-Us
- ISPs-R-Us now advertises a more specific route to Organization 1



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Hierarchical addressing: more specific routes

- Organization 1 moves from Fly-By-Night-ISP to ISPs-R-Us
- ISPs-R-Us now advertises a more specific route to Organization 1



That's actually **two** questions:

1. Q: How does a *host* get IP address within its network (host part of address)?
2. Q: How does a *network* get IP address for itself (network part of address)

How does *host* get IP address?

- hard-coded by sysadmin in config file (e.g., /etc/rc.config in UNIX)
- **DHCP**: Dynamic Host Configuration Protocol: dynamically get address from as server
 - “plug-and-play”

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DHCP: Dynamic Host Configuration Protocol

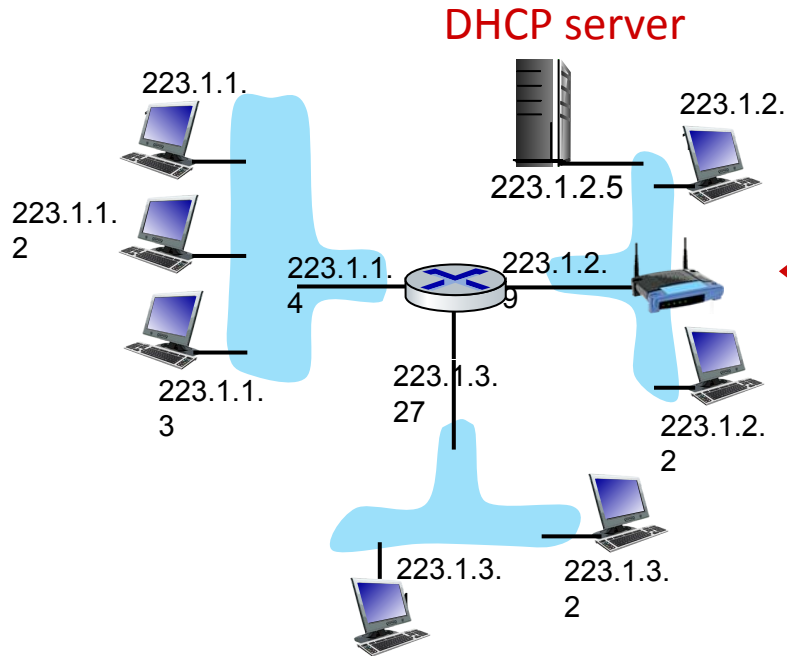


goal: host *dynamically* obtains IP address from network server when it “joins” network

- can renew its lease on address in use
- allows reuse of addresses (only hold address while connected/on)
- support for mobile users who join/leave network

DHCP overview:

- host broadcasts **DHCP discover** msg [optional]
- DHCP server responds with **DHCP offer** msg [optional]
- host requests IP address: **DHCP request** msg
- DHCP server sends address: **DHCP ack** msg



Typically, DHCP server will be co-located in router, serving all subnets to which router is attached



arriving **DHCP client** needs address in this network

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DHCP client-server scenario

DHCP server:
223.1.2.5



DHCP discover

Broadcast: is there a
DHCP server out there?

Arriving client



DHCP offer

Broadcast: I'm a DHCP
server! Here's an IP
address you can use

DHCP request

Broadcast: OK. I would
like to use this IP address!

DHCP ACK

Broadcast: OK. You've
got that IP address!

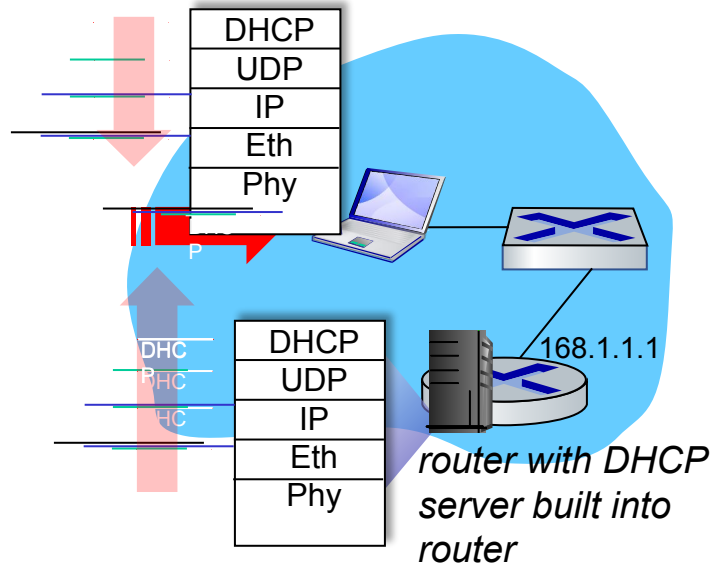
The two steps above can
be skipped "if a client
remembers and wishes to
reuse a previously
allocated network address"
[RFC 2131]

DHCP can return more than just allocated IP address on subnet:

- address of first-hop router for client
- name and IP address of DNS sever
- network mask (indicating network versus host portion of address)

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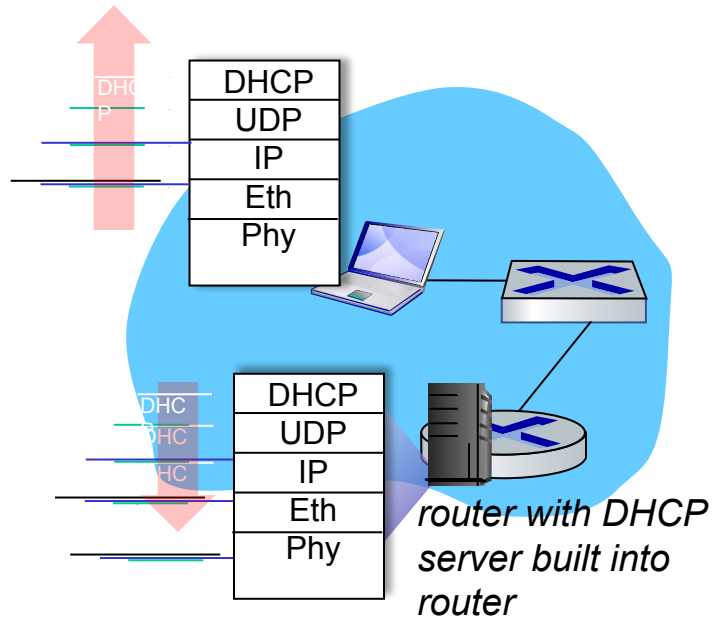
DHCP: example



- Connecting laptop will use DHCP to get IP address, address of first-hop router, address of DNS server.
- DHCP REQUEST message encapsulated in UDP, encapsulated in IP, encapsulated in Ethernet
- Ethernet frame broadcast (dest: FFFFFFFFFFFFFFFF) on LAN, received at router running DHCP server
- Ethernet demux'ed to IP demux'ed, UDP demux'ed to DHCP

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DHCP: example



- DCP server formulates DHCP ACK containing client's IP address, IP address of first-hop router for client, name & IP address of DNS server
- encapsulated DHCP server reply forwarded to client, demuxing up to DHCP at client
- client now knows its IP address, name and IP address of DNS server, IP address of its first-hop router

Q: how does an ISP get block of addresses?

A: ICANN: Internet Corporation for Assigned Names and Numbers

<http://www.icann.org/>

- allocates IP addresses, through 5 regional registries (RRs) (who may then allocate to local registries)
- manages DNS root zone, including delegation of individual TLD (.com, .edu, ...) management

Q: are there enough 32-bit IP addresses?

- ICANN allocated last chunk of IPv4 addresses to RRs in 2011
- NAT (next) helps IPv4 address space exhaustion
- IPv6 has 128-bit address space

"Who the hell knew how much address space we needed?" Vint Cerf (reflecting on decision to make IPv4 address 32 bits long)



THANK YOU

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